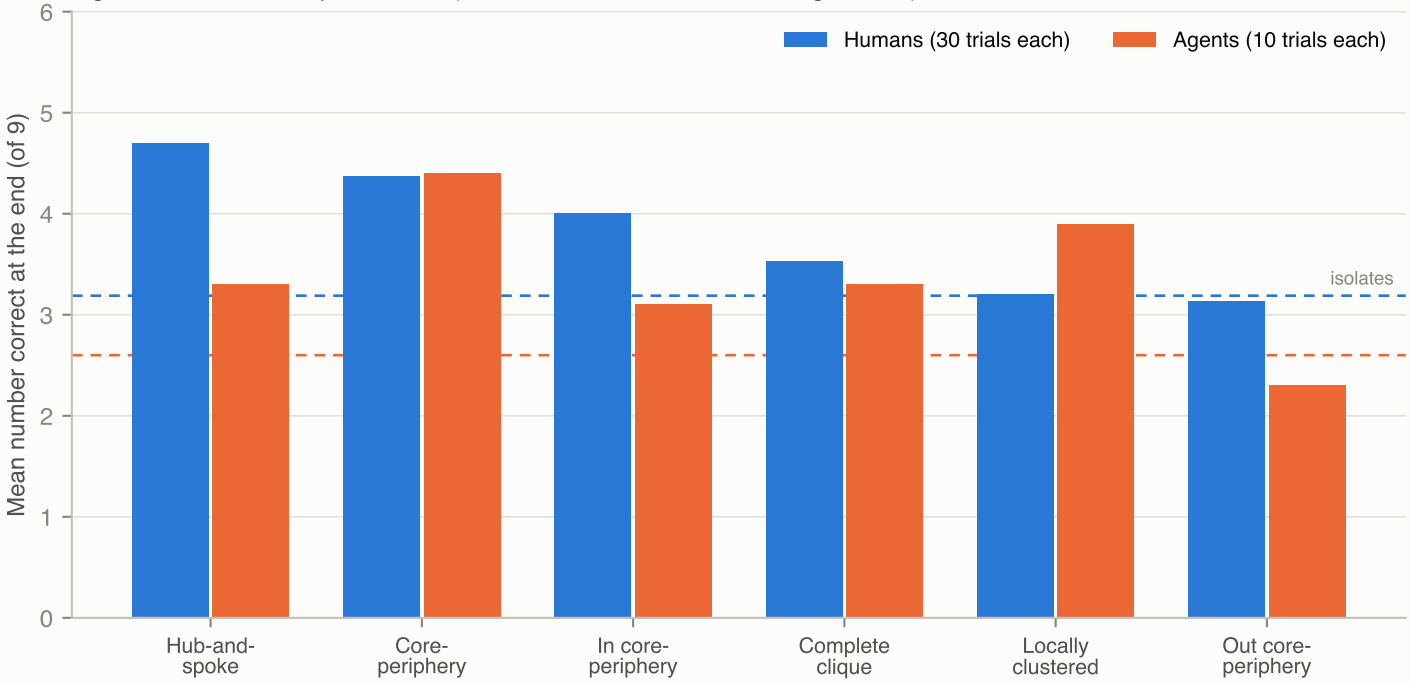
Network centralization with LLM agents

Replication of Bernstein, Shore & Jang (2023) with synthetic participants: final setting, 10 trials per network

Correct at the end (of 9)	Centralised minus controls	Out core-periphery	Spokes adopting correct hub
3.4 matches	**+0.25** too weak	**2.3** worst, matches	**41%** too low
humans 3.8	humans +1.17	humans 3.13	humans 64%

Figure 1. Mean correct per network (dashed: isolates, humans 3.19, agents 2.6)



Network	Mean, humans	Mean, agents	0–1 correct, h.	0–1 correct, a.	> 6 correct, h.	> 6 correct, a.
Hub-and-spoke	4.70	3.3	10%	30%	27%	20%
Core-periphery	4.37	4.4	27%	40%	33%	50%
In core-periphery	4.00	3.1	10%	30%	13%	20%
Complete clique	3.53	3.3	53%	60%	33%	40%
Locally clustered	3.20	3.9	33%	30%	10%	20%
Out core-periphery	3.13	2.3	30%	70%	13%	20%
All networks	3.82	3.4	27%	43%	22%	28%

Overall level, the complete clique, core-periphery and the weak out core-periphery match the human data. The main centralisation advantage is small and not significant (one-sided permutation p = 0.43), because hub-and-spoke is too low.

Distributions and hypotheses

Share of trials with at most k of 9 correct, as in the paper's Figure 6

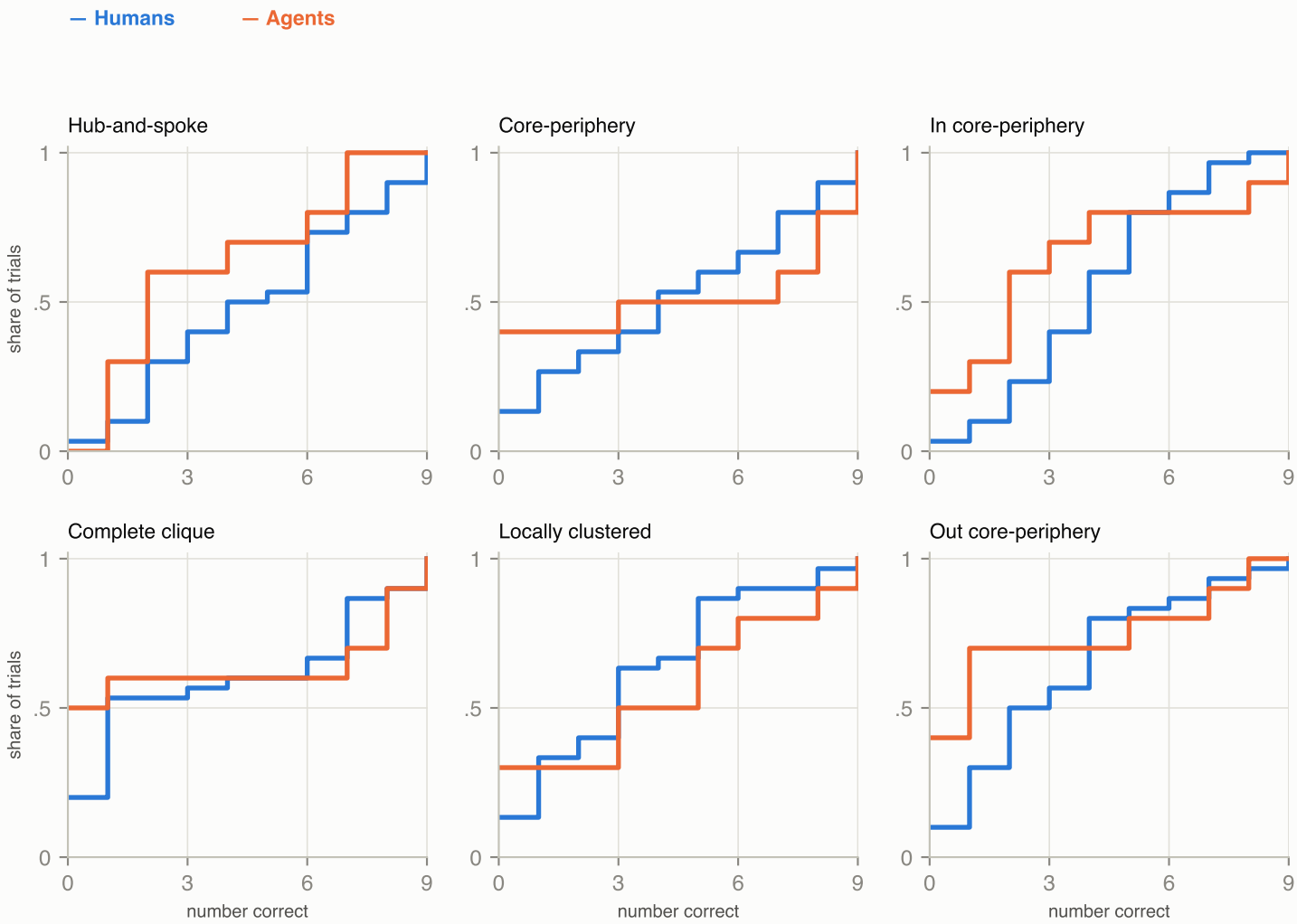


Figure 2. A curve further right means more participants correct. A steep jump at 0 or 9 means trials end all wrong or all right.

Table 2. The paper's hypotheses and mechanism checks, humans against the final agent setting (CP: core-periphery)

Hypothesis or mechanism	Humans	Agents	Replicates
H1 Fewer stuck trials (0–1 correct)	hub 10%, CP 27%, isolates 10%	hub 30%, CP 40%, isolates 30%	Partly
H2 Answer spreads (> 6 correct)	hub 27%, CP 33%, clique 33%	hub 20%, CP 50%, clique 40%	Partly
H3 Better on average (centralised minus controls)	+1.17	+0.25 (p = 0.43)	No
H4 One-way centralisation brings no benefit	out core-periphery 3.13 (worst)	2.3 (worst)	Yes
Mechanism: spokes adopt a correct hub	64%	41% (13 of 32)	No
Mechanism: out core-periphery core correct	32%	30% (9 of 30)	Yes
Answer changes matching a neighbour	65%	87%	No

How the settings moved the result

The main design variants tested; isolates were checked before every network run

Figure 3. The main contrast across settings

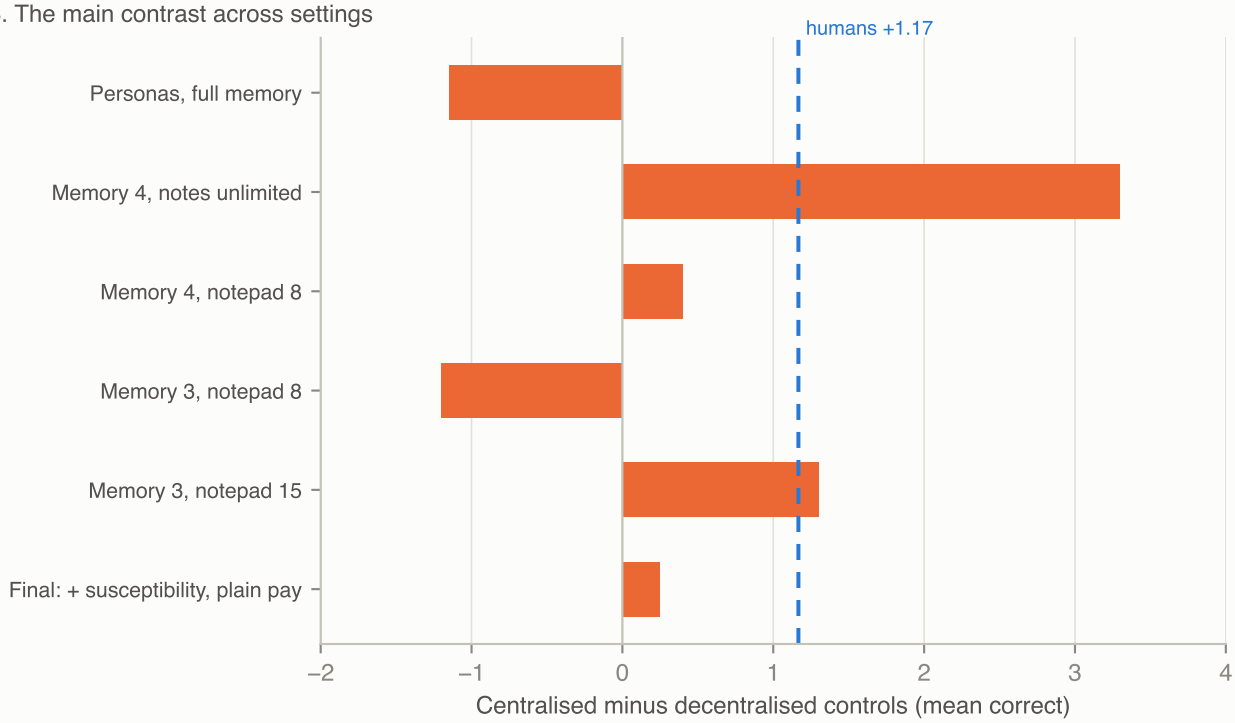


Table 3. Isolates and networks: mean correct of 9; 0–1 and > 6: share of network trials

Setting	Trials/net.	Isolates	Networks	0–1	> 6	Contrast	Hub followed
Humans	30	3.2	3.8	27%	22%	+1.17	64%
Personas, full memory	10	4.0	5.5	20%	45%	-1.15	33%
Memory 4, notes unlimited	5	2.7	5.5	27%	53%	+3.30	73%
Memory 4, notepad 8	5	2.4	1.8	63%	13%	+0.40	35%
Memory 3, notepad 8	5	2.7	1.8	63%	10%	-1.20	11%
Memory 3, notepad 15	5	2.9	4.9	27%	43%	+1.30	58%
Final: + susceptibility, plain pay	10	2.6	3.4	43%	28%	+0.25	41%

Full memory makes agents pool clues, so dense networks win. Limiting memory to the last few turns brings back learning from others' answers, but a notepad of 8 words makes every network lock onto the red herring. Memory 3 with a 15-word notepad sits closest; the susceptibility trait changed nothing measurable.

Setup and departures from the original

Element	Final setting
Task	The paper's murder mystery, 41 clues and three broadcasts (Appendix B.3); names redrawn per trial
Networks	Six treatments, 9 players; reconstructed from Appendix E and checked against the authors' data
Isolates	Nine unconnected agents (humans: resampled from in core-periphery nodes 5, 7, 9)
Time	15 minutes as 30 turns of 30 seconds; all agents act at once
Search	1 search per turn; a search can return a clue already found
Messages	Along network ties only; writing a message uses the turn's search
Agents	gpt-6-luna, reasoning effort drawn per agent (low, medium, high)
Personas	MTurk demographics (Levay et al. 2016), Big Five, susceptibility to influence
Memory	Last 3 turns plus a notepad of up to 15 words per turn, always visible
Calibration	Settings fixed on isolates only, before the network runs

Departures to report

Departure	What it means
Keywords per clue	Not published; rebuilt from the names in each clue
Discrete turns	Real time replaced by 30-second turns
Memory and notepad	Humans could scroll back through all messages; the limit models working memory
No dropout	16% of human trials had some dropout
Practice	Stage 1 only, no screening; stage 2 not run
Same model	All agents share one model; outcomes stay more bimodal than human ones

Cost: $28.30 for 424 trial records across calibration, pilots and the final run (gpt-6-luna list prices).
For comparison, the human study paid about $3,800 to its main-trial participants.